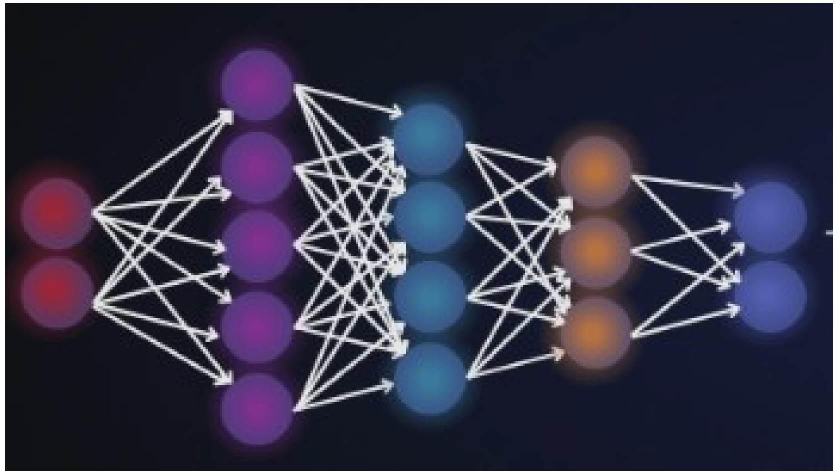
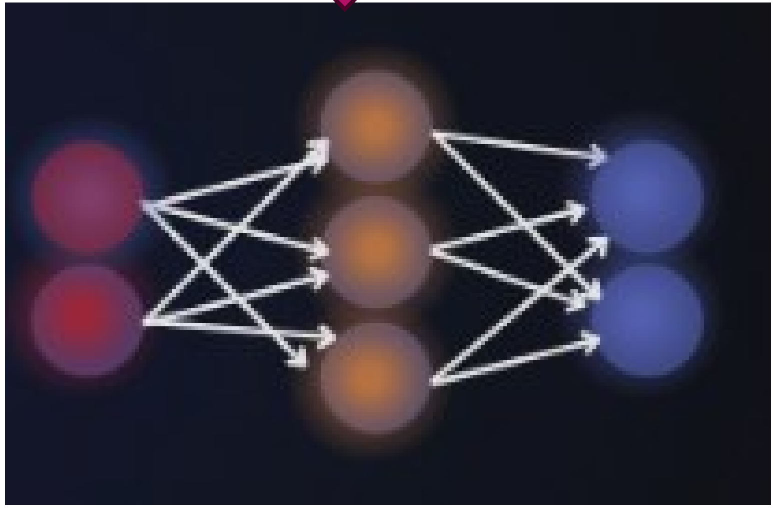
Model Optimization Techniques





Why Optimization is Needed

- 🧠 Deep learning models are resource intensive
  ⚙ Edge devices have limited memory & compute

What Optimization Achieves

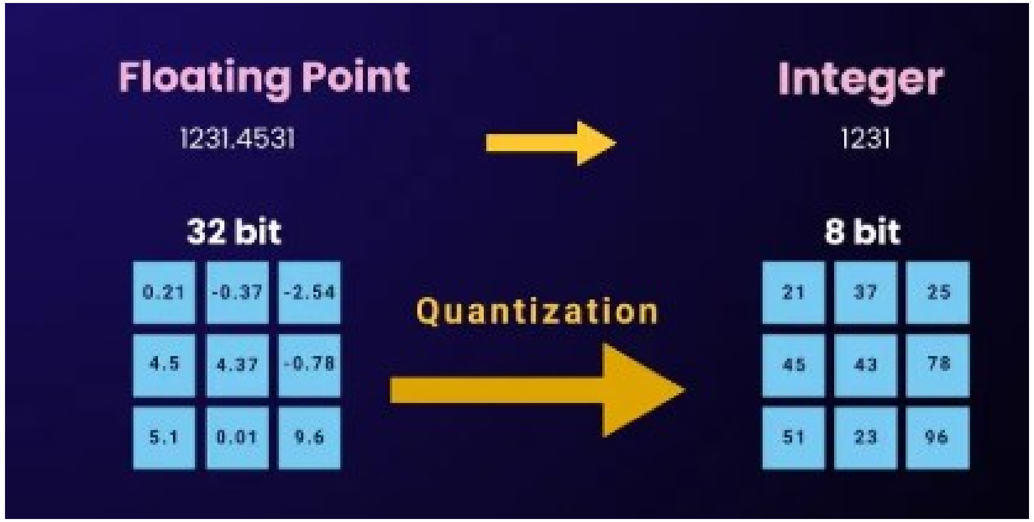
- 📉 Reduces model size & memory usage
  ⚡ Lowers compute requirements
  🔋 Improvs speed & energy efficiency

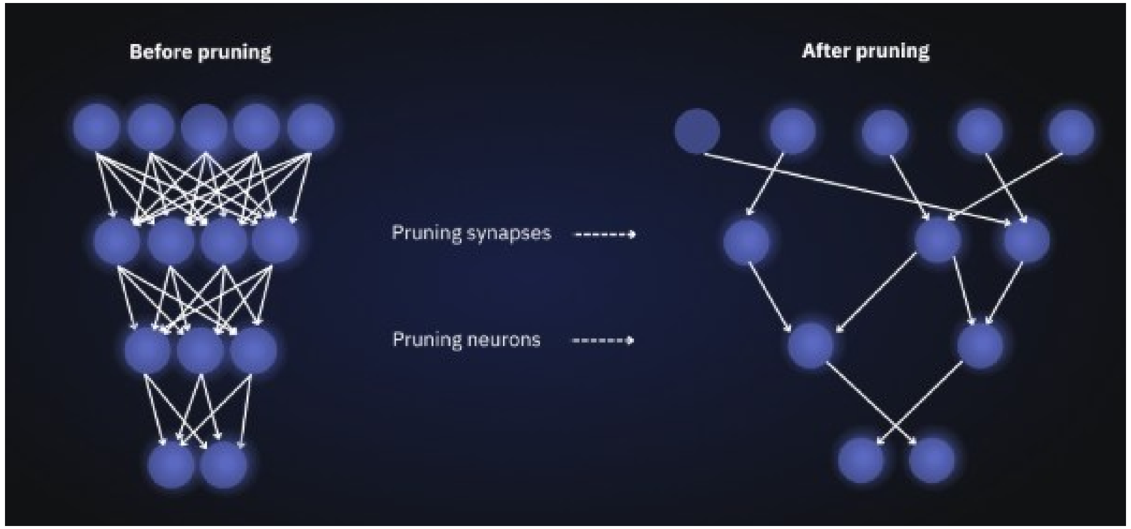
Why It Matters in AIoT

- ✔ Enables deployment on edge devices
  ✔ Maintains performance with minimal accuracy loss
  ✔ Turns lab models into real-world solutions

Optimization makes AI models smaller, faster, and edge-ready

Key AIoT Model Optimization Techniques





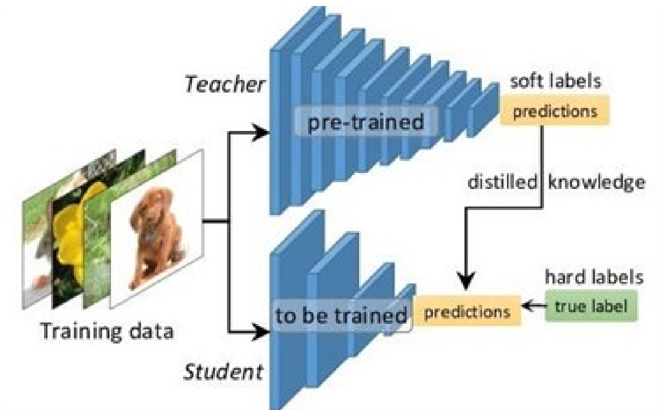
Quantization

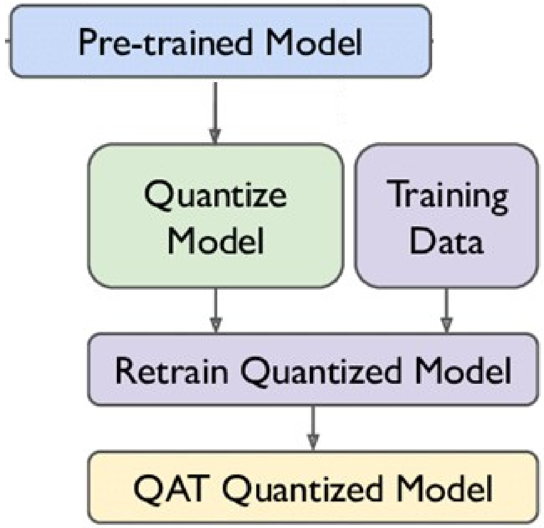
- Convert 32-bit precision → 8-bit / 16-bit integers
  ✓ Smaller model & faster inference
  ✓ Lower power consumption
- **Example:** A smart camera uses an 8-bit quantized object detection model to identify intruders in real time without needing cloud processing.

Pruning

- Remove unnecessary weights & neurons
  ✓ Reduces compute & memory usage
- **Structured Pruning** – remove filters/layers (hardware friendly)
- **Unstructured Pruning** – remove individual weights
- **Example:** A predictive maintenance model removes redundant neurons to run efficiently on an industrial gateway.

Advanced Model Optimization Techniques for Edge AI





Knowledge Distillation

- A smaller **student model** learns to mimic a larger **teacher model**
- Produces compact models with similar performance
- Reduces compute and memory requirements

Example:

- Healthcare wearables use **distilled models** to detect abnormal heart rhythms using lightweight AI running directly on the device.

Quantization-Aware Training

- Simulates **low-precision computation during training**
- Helps the model adapt to reduced precision
- Maintains higher accuracy after deployment

Example:

- Smart retail cameras trained with quantization-aware techniques maintain high detection accuracy when running on **low-power edge hardware**.

Architecture Optimization for Edge AI

Architecture Optimization : Instead of compressing large models, developers design **efficient architectures specifically for edge devices.**

Examples of Edge-Optimized Models:

- **MobileNet**
- **EfficientNet-Lite**

These models are designed to:

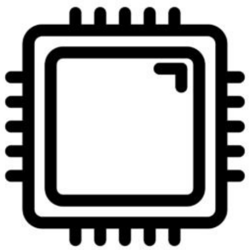
- Reduce computational complexity
- Minimize memory usage
- Enable real-time inference on edge hardware

Example:

- MobileNet powers **face detection in smartphones and smart doorbells**, delivering strong performance with minimal resources.

Optimization techniques such as **distillation, quantization-aware training, and efficient architectures** transform large AI models into **edge-ready intelligence**.

Benefits for AIoT Devices



🧠 Lower Memory Usage

- Requires less RAM
- Improves stability & performance on constrained devices

⚡ Faster Inference

- Fewer calculations & lighter models
- Enables real-time response & low latency

🔋 Energy Efficiency

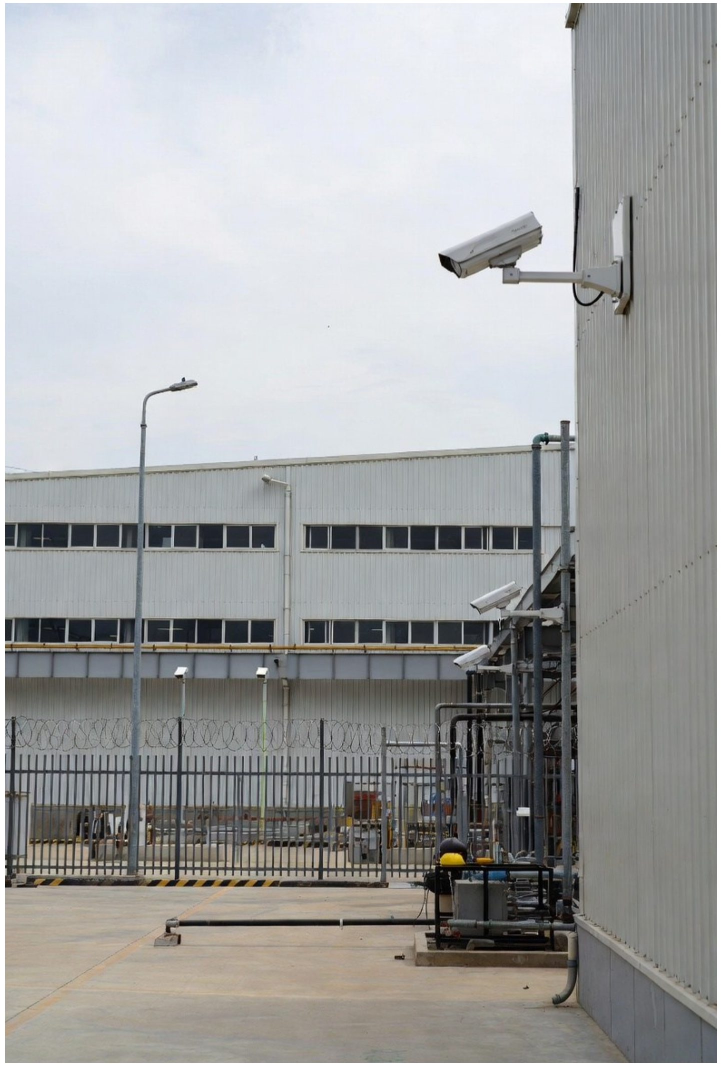
- Reduced compute workload
- Extends battery life in sensors & wearables

💾 Reduced Storage Requirements

- Smaller model footprint
- Enables deployment on limited storage devices

Optimization enables efficient, reliable, and real-time AI at the edge

Case Study: Optimizing an AI Model for Smart Factory Safety



Background

- A manufacturing plant installed AI-powered cameras to detect whether workers were wearing safety helmets and protective gear. The original deep learning model was accurate but required cloud processing, causing delays and high bandwidth usage.

Challenges

- High latency when sending video to the cloud
- Bandwidth costs due to continuous video streaming
- Privacy concerns with transmitting worker footage
- Edge gateway struggled to run the full model

Case Study: Optimizing an AI Model for Smart Factory Safety

Optimization Approach

1 Quantization
The model was converted from 32-bit to 8-bit precision. ➡ Reduced model size by ~75% and improved processing speed.

2 Pruning
Redundant network weights were removed. ➡ Reduced computational load and improved efficiency.

3 Knowledge Distillation
A smaller "student" model was trained from the original model. ➡ Maintained accuracy while reducing complexity.

4 Edge Deployment
The optimized model was deployed on an edge gateway connected to cameras.

Results

✓ Helmet detection performed in real time (< 100 ms)

✓ 80% reduction in bandwidth usage

✓ Worker privacy preserved (no cloud video transfer)

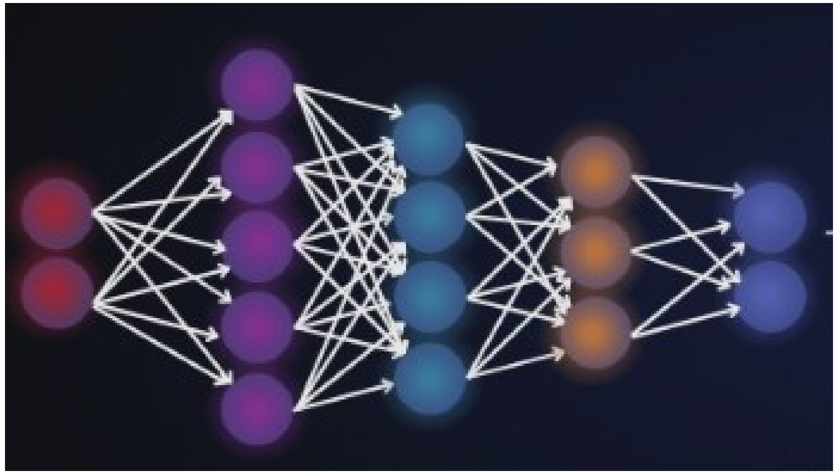
✓ Edge device power consumption reduced significantly

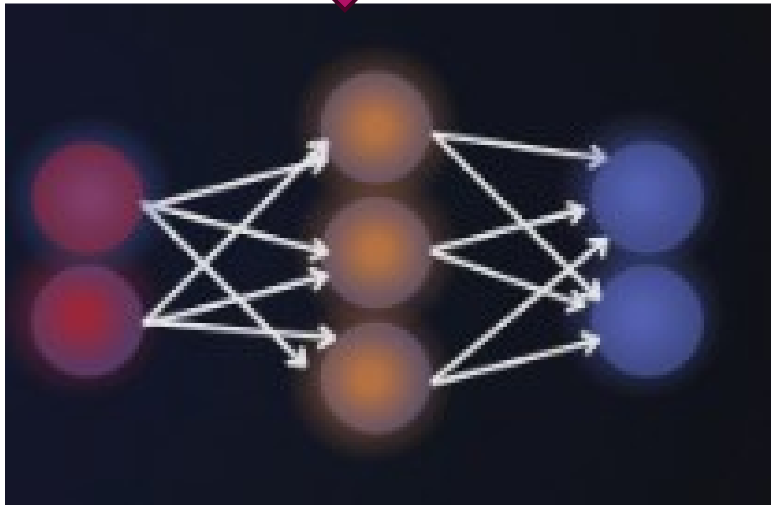
✓ Safety violations detected instantly

Business Impact

- Improved worker safety compliance
- Reduced operational costs
- Faster incident response
- Enabled scalable deployment across facilities

Model Optimization Techniques





Key Learning Takeaways

✓ Optimization enables AI models to run on constrained edge devices

✓ Quantization and pruning improve speed and efficiency

✓ Knowledge distillation preserves performance while reducing size

✓ Edge deployment enhances privacy, speed, and cost efficiency